Md. Motiur Rahman



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Career Objective

A dedicated and results-driven GIS and Remote Sensing specialist with above 7 years of professional experience. Seeking a challenging role that will leverage my expertise in GIS, Remote Sensing, spatial modeling, and data analysis to deliver sustainable solutions. Passionate about collaborating with interdisciplinary teams to utilize spatial data and innovative technology for informed decision-making, resource management, and environmental conservation.

Technical Skills

- **Geospatial Analysis & GIS Software:** Proficient in Google Earth Engine (GEE), ArcGIS, ArcGIS Pro, QGIS, ERDAS IMAGINE, and ENVI for advanced spatial analysis, geoprocessing, cartographic mapping, and geospatial data visualization.
- **Geospatial Data Collection & Management:** Experienced in field data collection, GNSS/GPS and RTK surveying, spatial database development, geospatial data processing, and mobile data collection platforms, including KoboToolbox and ODK.
- **Drone Operations & UAV Piloting:** Skilled in UAV flight planning, mission design, autonomous flight operations, aerial data acquisition, and flight quality control using DJI Pilot, DJI GS Pro, Pix4Dcapture, and other UAV flight control applications.
- **UAV Photogrammetry & 3D Mapping:** Experienced in processing UAV aerial imagery using Agisoft Metashape Professional, Pix4D, DJI Terra, and Correlator3D to generate high-resolution orthomosaics, Digital Surface Models (DSM), Digital Terrain Models (DTM), dense point clouds, contour maps, and 3D models.
- **Remote Sensing & Image Analysis:** Skilled in satellite and multispectral imagery processing, vegetation assessment, NDVI generation, land use/land cover (LULC) classification, change detection, and agricultural monitoring.
- **Programming & Automation:** Basic proficiency in Python and Java for geospatial data processing, spatial analysis, and workflow automation.
- **CAD & Engineering Design:** Basic knowledge of AutoCAD and Civil 3D, including 2D drafting, basic drawing editing, topographic drawing interpretation, and contour mapping.
- **Microsoft Office & Data Management:** Advanced proficiency in Microsoft Word, Excel, PowerPoint, and Access for technical documentation, data analysis, reporting, and presentations.

Professional Experience

UAV & Photogrammetry Expert

Geospatial Works — Dhaka, Bangladesh

April 2026 – Present

Project: Research Project on Irrigation Water Management and Web-based Agricultural Information System for Bangladesh (WEB-AIS)

Responsibilities:

- Plan and conduct UAV-based aerial surveys, including flight mission planning, data acquisition, and field operations using professional surveying drones.
- Process UAV imagery using photogrammetric software to generate high-resolution orthomosaics, Digital Surface Models (DSM), Digital Terrain Models (DTM), and 3D point clouds.
- Conduct high-precision GNSS/RTK surveys for ground control, georeferencing, and spatial data collection.
- Perform GIS data management, spatial analysis, geospatial database preparation, and thematic mapping.
- Apply remote sensing techniques for agricultural monitoring, land-use analysis, and vegetation assessment.
- Generate and analyze Normalized Difference Vegetation Index (NDVI) maps using multispectral UAV imagery.
- Extract agricultural and physical features, field boundaries, irrigation infrastructure, and other spatial information from high-resolution orthomosaic imagery.
- Conduct spatial data extraction, interpretation, validation, and quality assessment to ensure accuracy and reliability.
- Prepare GIS maps, survey outputs, geospatial datasets, and technical documentation to support irrigation water management and agricultural information systems.
- Integrate UAV, GNSS/RTK, GIS, and remote sensing datasets to support research activities and web-based agricultural information services.

GIS Expert

Asian Development Bank (ADB) — Dhaka, Bangladesh

August 2025 – March 2026

Project: Economic Corridor Study.

Responsibilities:

- Compiled and managed corridor-wide geospatial datasets, including transport infrastructure, water systems, ecological zones, human settlements, and land use.
- Ensured data accuracy, consistency, and interoperability across multiple GIS platforms.

- Developed thematic and layered GIS maps (flood zones, connectivity, ecological sensitivity, and socio-economics) to support planning and decision-making.
- Designed and maintained a centralized GIS platform enabling multi-sectoral integration and real-time spatial analysis.
- Produced GIS overlays to identify intersections and conflicts (e.g., infrastructure vs. natural water flow).
- Applied GIS tools to analyze flood-prone areas, siltation risks, and vulnerable ecological and human settlements.
- Supported scenario modeling for climate change impacts and resilience planning.
- Mapped cross-border linkages (Nepal, Bhutan, India, Myanmar) to visualize trade routes, transport corridors, and integration opportunities.
- Collaborated with transport, water, environment, and economic experts to provide actionable spatial insights.
- Prepared GIS-based reports, dashboards, and visual presentations for government, communities, and development partners.
- Conducted GIS training for stakeholders to strengthen spatial decision-making capacity.
- Contributed spatial evidence and recommendations to the master plan for sustainable corridor development.

GIS Specialist

EQMS Consulting Limited — Dhaka, Bangladesh
May 2025 – July 2025
Project: Feasibility Study and Developing a Master Plan for Bhawal National Park.

Responsibilities:

- Acquired, processed, and analyzed spatial data using advanced GIS and Remote Sensing tools.
- Designed, developed, and maintained geospatial databases to support the Bhawal National Park Master Plan.
- Conducted spatial analyses and produced thematic maps (e.g., land use, habitat distribution, vegetation cover).
- Integrated field data from RTK and UAV surveys with satellite imagery for enhanced spatial modeling and interpretation.
- Performed georeferencing, digitization, spatial adjustment, and edge matching of scanned Mouza maps.
- Generated high-resolution Orthomosaics, digital elevation models (DEMs/DTMs), and vegetation indices using photogrammetry software.

- Conducted field validation and accuracy assessments to ensure reliability of remote sensing-based outputs.
- Prepared comprehensive GIS reports, metadata documentation, and visual maps for technical and planning deliverables.
- Collaborated with multidisciplinary teams, including ecologists and planners, to provide spatial insights for sustainable infrastructure and forest management.

Remote Sensing and GIS Specialist

Asian Development Bank (ADB) — Dhaka, Bangladesh
October 2023 – March 2025
Project: Using Frontier Technology and Big Data Analytics for Smart Infrastructure Facility Planning and Monitoring

Responsibilities:

- Collected high-resolution satellite (Sentinel-1, Sentinel-2) and UAV (DJI Mavic 3M with RTK) imagery for agricultural and flood monitoring.
- Conducted field surveys using RTK/GPS and iCrops mobile app for ground truth data.
- Processed UAV data to generate orthomosaics, DEMs, and vegetation indices (NDVI, EVI) using DJI Terra, Pix4D, and GEE.
- Performed radiometric, geometric, and atmospheric corrections on satellite imagery.
- Applied machine learning (e.g., Random Forest, SVM) for crop type classification and area estimation.
- Conducted land use change detection and flood impact assessment.
- Created spatial distribution maps and damage assessment layers using ArcGIS Pro and QGIS.
- Produced thematic maps and dashboards for stakeholder reporting.
- Integrated remote sensing outputs with field data and historical yield records.
- Supported M&E through geostatistical analyses and evidence-based reporting for smart infrastructure planning.

Assistant Consultant, GIS and Remote Sensing Department

EQMS Consulting Limited — Dhaka, Bangladesh
April 2022 – September 2023
Projects: GIS and Remote Sensing, Feasibility Studies, Environmental Impact Assessments

Responsibilities:

- Conducted aerial surveys using UAVs and collected precise field data with GPS, GNSS, and RTK systems.

- Used mobile data platforms (e.g., Kobo Toolbox, ODK, CTO) for field data collection.
- Acquired satellite imagery and existing geospatial datasets from various sources.
- Integrated field data with existing spatial databases to create comprehensive geospatial datasets.
- Performed data cleaning, error correction, and optimized data structures for spatial analysis and ensured consistency, accuracy, and usability of spatial datasets across projects.
- Applied photogrammetric techniques for elevation modeling, surface analysis, settlement mapping, and change detection.
- Supported project planning and monitoring through spatial data analysis and field survey planning.
- Assisted in low-income settlement design using AutoCAD and GIS tools.
- Contributed to project proposals, including technical writing and methodology development.
- Coordinated with clients, subcontractors, and multidisciplinary teams for quality assurance and data validation.
- Prepared technical reports, GIS deliverables, and visual materials for stakeholder presentations.
- Verified data accuracy through field validation and ground-truthing surveys.

Research Consultant, Remote Sensing Division

Center for Environmental and Geographic Information Services (CEGIS) — Dhaka, Bangladesh
June 2019 – March 2022

Projects: Urban & rural settlement mapping for the Population Census 2021, BPDB Master Plan, Coastal Embankment Improvement, Salinization Mapping, and Canal Rehabilitation

Responsibilities:

- Designed and managed national-scale spatial databases integrating mouza maps, satellite imagery, historical maps, and field survey data.
- Supervised data compilation teams and implemented quality control measures to ensure spatial accuracy and consistency.
- Conducted multi-temporal satellite image analysis for flood zone mapping, land use/land cover change detection.
- Mapped salinity-affected zones, waterlogged areas, and coastal vulnerability using tools such as ERDAS Imagine, ENVI, and ArcGIS.
- Performed accuracy assessments by comparing classified outputs with ground truth observations.
- Led field missions for data verification, site observations, and validation of satellite-derived outputs.

- Coordinated with local authorities to identify waterlogging hotspots and encroachment issues aligned with mouza boundaries (e.g., in Chattogram City).
- Created detailed thematic maps and data visualizations tailored for planners, policymakers, and project stakeholders.
- Applied classification techniques and symbology standards to produce high-quality cartographic outputs.
- Drafted technical reports presenting analysis findings to support environmental compliance, social safeguard implementation, and infrastructure planning.

Education

M.Sc. in Geography and Environment
Jahangirnagar University, Bangladesh — 2018
GPA: 3.60/4.00
B.Sc. in Geography and Environment
Jahangirnagar University, Bangladesh — 2017
CGPA: 3.57/4.00

Certifications & Training

- Training on *"Use of Drone and Satellite Images for Crop Monitoring and Crop Damage Assessment"*
  Asian Development Bank (ADB) & ICRISAT - April 2024
- *"The Systems, Flight, and Required Safety Training"* for Drones in Climate Smart Agriculture and Water Management
  AIRON of the Republic of Korea - March-April 2024
- *"The Data Processing and Analytics Production Training"* for Drones in Climate Smart Agriculture and Water Management
  Hojung Solutions, Republic of Korea — March-April 2024
- Training on *"Satellite Meteorology Applications in Weather and Climate Studies"*
  Indian Institute of Remote Sensing (IIRS), India — August 2023
- Training on *"Fundamental GIS and Remote Sensing"*
  Jahangirnagar University, Bangladesh — January 2018
- *"Non-credit computer course in windows operating system, Microsoft office including word, excel, and power point"*
  Jahangirnagar University, Dhaka, Bangladesh - March - October 2015

Publications

- **Biswas, R. N., et al.** (2022). Impacts of morphological change on coastal landscape dynamics in Monpura Island in the northern Bay of Bengal, Bangladesh. *Regional Studies in Marine Science, 53*, 102403. DOI: 10.1016/j.rsma.2022.102403